## 3.4 Linear inequalities

You can solve linear inequalities using similar methods to those for solving linear equations.

- **The solution of an inequality is the set of all real numbers $x$ that make the inequality true.**

### Example 7

Find the set of values of $x$ for which:

**a** $5x + 9 \geqslant x + 20$ **b** $12 - 3x < 27$

**c** $3(x - 5) > 5 - 2(x - 8)$

**a** $5x + 9 \geqslant x + 20$

$4x + 9 \geqslant 20$

$4x \geqslant 11$

$x \geqslant 2.75$

Rearrange to get $x \geqslant \ldots$

**b** $12 - 3x < 27$

Subtract 12 from both sides.

$-3x < 15$

Divide both sides by $-3$. (You therefore need to turn round the inequality sign.)

$x > -5$

In set notation $\{x : x > -5\}$.

**c** $3(x - 5) > 5 - 2(x - 8)$

$3x - 15 > 5 - 2x + 16$

Multiply out (note: $-2 \times -8 = +16$).

$5x > 5 + 16 + 15$

Rearrange to get $x > \ldots$

$5x > 36$

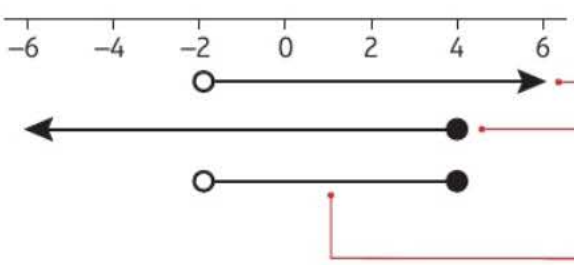
$x > 7.2$

In set notation $\{x : x > 7.2\}$.

**Notation** You can write the solution to this inequality using set notation as $\{x : x \geqslant 2.75\}$. This means the set of all values $x$ for which $x$ is greater than or equal to 2.75.

You may sometimes need to find the set of values for which **two** inequalities are true together. Number lines can be useful to find your solution.

For example, in the number line below the solution set is $x > -2$ **and** $x \leqslant 4$.

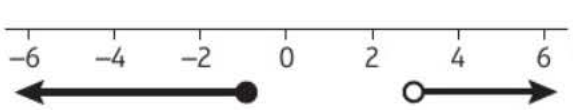



○ is used for $<$ and $>$ and means the end value is not included.

● is used for $\leqslant$ and $\geqslant$ and means the end value is included.

These are the only real values that satisfy both equalities simultaneously so the solution is $-2 < x \leqslant 4$.

Here the solution sets are $x \leqslant -1$ **or** $x > 3$.




Here there is no overlap and the two inequalities have to be written separately as $x \leqslant -1$ or $x > 3$.

**Notation** In set notation

$x > -2$ **and** $x \leqslant 4$ is written $\{x : -2 < x \leqslant 4\}$

or alternatively $\{x : x > -2\} \cap \{x : x \leqslant 4\}$

$x \leqslant -1$ **or** $x > 3$ is written $\{x : x \leqslant -1\} \cup \{x : x > 3\}$

## Example 8

Find the set of values of $x$ for which:

**a** $3x - 5 < x + 8$ and $5x > x - 8$

**b** $x - 5 > 1 - x$ or $15 - 3x > 5 + 2x$.

**c** $4x + 7 > 3$ and $17 < 11 + 2x$.

**a** $3x - 5 < x + 8$ $\qquad$ $5x > x - 8$

$2x - 5 < 8$ $\qquad$ $4x > -8$

$2x < 13$ $\qquad$ $x > -2$

$x < 6.5$




So the required set of values is $-2 < x < 6.5$.

Draw a number line to illustrate the two inequalities.

The two sets of values overlap (intersect) where $-2 < x < 6.5$.

Notice here how this is written when $x$ lies between two values.

In set notation this can be written as $\{x : -2 < x < 6.5\}$.

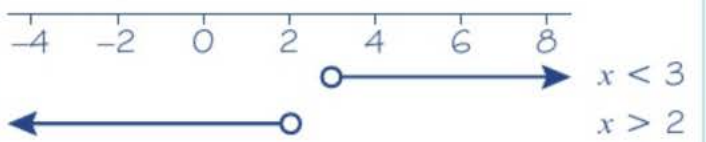
**b** $x - 5 > 1 - x$ $\qquad$ $15 - 3x > 5 + 2x$

$2x - 5 > 1$ $\qquad$ $10 - 3x > 2x$

$2x > 6$ $\qquad$ $10 > 5x$

$x > 3$ $\qquad$ $2 > x$

$x < 2$




The solution is $x > 3$ or $x < 2$.

Draw a number line. Note that there is no overlap between the two sets of values.

In set notation this can be written as $\{x : x < 2\} \cup \{x : x > 3\}$.

## Exercise 3D

**1** Find the set of values of $x$ for which:

**a** $2x - 3 < 5$

**b** $5x + 4 \geqslant 39$

**c** $6x - 3 > 2x + 7$

**d** $5x + 6 \leqslant -12 - x$

**e** $15 - x > 4$

**f** $21 - 2x > 8 + 3x$

**g** $1 + x < 25 + 3x$

**h** $7x - 7 < 7 - 7x$

**i** $5 - 0.5x \geqslant 1$

**j** $5x + 4 > 12 - 2x$

**2** Find the set of values of $x$ for which:

**a** $2(x-3) \geqslant 0$ **b** $8(1-x) > x-1$ **c** $3(x+7) \leqslant 8-x$

**d** $2(x-3)-(x+12) < 0$ **e** $1+11(2-x) < 10(x-4)$ **f** $2(x-5) \geqslant 3(4-x)$

**g** $12x-3(x-3) < 45$ **h** $x-2(5+2x) < 11$ **i** $x(x-4) \geqslant x^2+2$

**j** $x(5-x) \geqslant 3+x-x^2$ **k** $3x+2x(x-3) \leqslant 2(5+x^2)$ **l** $x(2x-5) \leqslant \dfrac{4x(x+3)}{2} - 9$

**3** Use set notation to describe the set of values of $x$ for which:

**a** $3(x-2) > x-4$ and $4x+12 > 2x+17$

**b** $2x-5 < x-1$ and $7(x+1) > 23-x$

**c** $2x-3 > 2$ and $3(x+2) < 12+x$

**d** $15-x < 2(11-x)$ and $5(3x-1) > 12x+19$

**e** $3x+8 \leqslant 20$ and $2(3x-7) \geqslant x+6$

**f** $5x+3 < 9$ or $5(2x+1) > 27$

**g** $4(3x+7) \leqslant 20$ or $2(3x-5) \geqslant \dfrac{7-6x}{2}$

**Challenge**

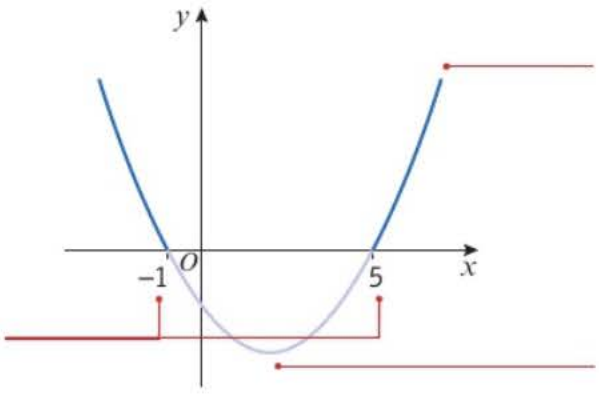
$A = \{x : 3x+5 > 2\}$ $B = \left\{x : \frac{x}{2}+1 \leqslant 3\right\}$ $C = \{x : 11 < 2x-1\}$

Given that $A \cap (B \cup C) = \{x : p < x \leqslant q\} \cup \{x : x > r\}$, find the values of $p$, $q$ and $r$.

## 3.5 Quadratic inequalities

- **To solve a quadratic inequality:**
  - **Rearrange so that the right-hand side of the inequality is 0**
  - **Solve the corresponding quadratic equation to find the critical values**
  - **Sketch the graph of the quadratic function**
  - **Use your sketch to find the required set of values.**

The sketch shows the graph of $f(x) = x^2 - 4x - 5$
$= (x+1)(x-5)$




The solutions to $f(x) = 0$ are $x = -1$ and $x = 5$. These are called the critical values.

The solutions to the quadratic inequality $x^2 - 4x - 5 > 0$ are the $x$-values when the curve is **above** the $x$-axis (the darker part of the curve). This is when $x < -1$ or $x > 5$. In set notation the solution is $\{x : x < -1\} \cup \{x : x > 5\}$.

The solutions to the quadratic inequality $x^2 - 4x - 5 < 0$ are the $x$-values when the curve is **below** the $x$-axis (the lighter part of the curve). This is when $x > -1$ and $x < 5$ or $-1 < x < 5$. In set notation the solution is $\{x : -1 < x < 5\}$.

## Example 9

Find the set of values of $x$ for which:

$3 - 5x - 2x^2 < 0$.

$3 - 5x - 2x^2 = 0$

Quadratic equation.

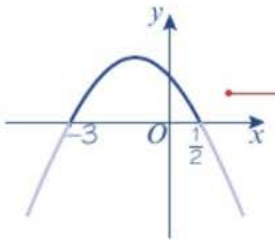
$2x^2 + 5x - 3 = 0$

Multiply by $-1$ (so it's easier to factorise).

$(2x - 1)(x + 3) = 0$

$x = \frac{1}{2}$ or $x = -3$

$\frac{1}{2}$ and $-3$ are the critical values.




Draw a sketch to show the shape of the graph and the critical values.

Since the coefficient of $x^2$ is negative, the graph is 'upside-down ∪-shaped'. It crosses the $x$-axis at $-3$ and $\frac{1}{2}$. ← Section 2.4

So the required set of values is

$x < -3$ or $x > \frac{1}{2}$.

$3 - 5x - 2x^2 < 0$ ($y < 0$) for the outer parts of the graph, below the $x$-axis, as shown by the paler parts of the curve.

In set notation this can be written as

$\{x : x < -3\} \cup \{x : x > \frac{1}{2}\}$.

## Example 10

**a** Find the set of values of $x$ for which $12 + 4x > x^2$.

**b** Hence find the set of values for which $12 + 4x > x^2$ and $5x - 3 > 2$.

**a** $12 + 4x > x^2$

$0 > x^2 - 4x - 12$

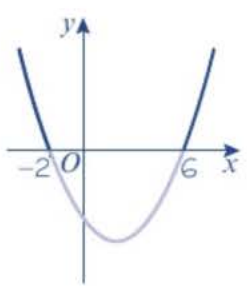
$x^2 - 4x - 12 < 0$

$x^2 - 4x - 12 = 0$

$(x + 2)(x - 6) = 0$

$x = -2$ or $x = 6$

Sketch of $y = x^2 - 4x - 12$




$x^2 - 4x - 12 < 0$

Solution: $-2 < x < 6$

You can use a table to check your solution.

$-2 < x < 6$

Use the critical values to split the real number line into sets.



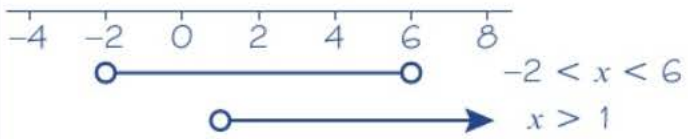

| | $x < -2$ | $-2 < x < 6$ | $x > 6$ |
|---|---|---|---|
| $x + 2$ | − | + | + |
| $x - 6$ | − | − | + |
| $(x + 2)(x - 6)$ | + | − | + |

For each set, check whether the set of values makes the value of the bracket positive or negative. For example, if $x < -2$, $(x + 2)$ is negative, $(x - 6)$ is negative, and $(x + 2)(x - 6)$ is (neg) × (neg) = positive.

In set notation the solution is $\{x : -2 < x < 6\}$.

**b** Solving $12 + 4x > x^2$ gives $-2 < x < 6$.

Solving $5x - 3 > 2$ gives $x > 1$.




The two sets of values overlap where $1 < x < 6$.

So the solution is $1 < x < 6$.

**Problem-solving**

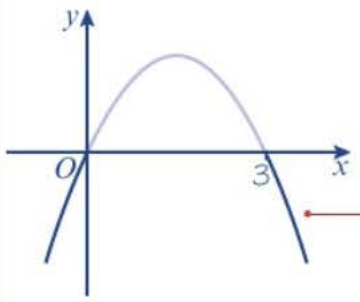
This question is easier if you represent the information in more than one way. Use a sketch graph to solve the quadratic inequality, and use a number line to combine it with the linear inequality.

In set notation this can be written as $\{x : 1 < x < 6\}$.

## Example 11

Find the set of values for which $\frac{6}{x} > 2$, $x \neq 0$

$$\frac{6}{x} > 2$$
$$6x > 2x^2$$
$$6x - 2x^2 > 0$$
$$6x - 2x^2 = 0$$
$$x(6 - 2x) = 0$$
$$x = 0 \text{ or } x = 3$$




The solution is $0 < x < 3$.

**Watch out** $x$ could be either positive or negative, so you can't multiply both sides of this inequality by $x$. Instead, multiply both sides by $x^2$. Because $x^2$ is never negative, and $x \neq 0$ so $x^2 \neq 0$, the inequality sign stays the same.

Solve the corresponding quadratic equation to find the critical values.

$x = 0$ can still be a critical value even though $x \neq 0$. But it would not be part of the solution set, even if the inequality was $\geqslant$ rather than $>$.

Sketch $y = x(6 - 2x)$. You are interested in the values of $x$ where the graph is above the $x$-axis.

In set notation this can be written as $\{x : 0 < x < 3\}$.

## Exercise 3E

**1** Find the set of values of $x$ for which:

**a** $x^2 - 11x + 24 < 0$ **b** $12 - x - x^2 > 0$ **c** $x^2 - 3x - 10 > 0$

**d** $x^2 + 7x + 12 \geqslant 0$ **e** $7 + 13x - 2x^2 > 0$ **f** $10 + x - 2x^2 < 0$

**g** $4x^2 - 8x + 3 \leqslant 0$ **h** $-2 + 7x - 3x^2 < 0$ **i** $x^2 - 9 < 0$

**j** $6x^2 + 11x - 10 > 0$ **k** $x^2 - 5x > 0$ **l** $2x^2 + 3x \leqslant 0$

**2** Find the set of values of $x$ for which:

**a** $x^2 < 10 - 3x$ **b** $11 < x^2 + 10$

**c** $x(3 - 2x) > 1$ **d** $x(x + 11) < 3(1 - x^2)$

**3** Use set notation to describe the set of values of $x$ for which:

**a** $x^2 - 7x + 10 < 0$ and $3x + 5 < 17$

**b** $x^2 - x - 6 > 0$ and $10 - 2x < 5$

**c** $4x^2 - 3x - 1 < 0$ and $4(x + 2) < 15 - (x + 7)$

**d** $2x^2 - x - 1 < 0$ and $14 < 3x - 2$

**e** $x^2 - x - 12 > 0$ and $3x + 17 > 2$

**f** $x^2 - 2x - 3 < 0$ and $x^2 - 3x + 2 > 0$

(P) **4** Given that $x \neq 0$, find the set of values of $x$ for which:

**a** $\frac{2}{x} < 1$

**b** $5 > \frac{4}{x}$

**c** $\frac{1}{x} + 3 > 2$

**d** $6 + \frac{5}{x} > \frac{8}{x}$

**e** $25 > \frac{1}{x^2}$

**f** $\frac{6}{x^2} + \frac{7}{x} \leqslant 3$

**5 a** Find the range of values of $k$ for which the equation $x^2 - kx + (k + 3) = 0$ has no real roots.

**b** Find the range of values of $p$ for which the roots of the equation $px^2 + px - 2 = 0$ are real.

**Hint** The quadratic equation $ax^2 + bx + c = 0$ has real roots if $b^2 - 4ac \geqslant 0$. ← Section 2.5

(E) **6** Find the set of values of $x$ for which $x^2 - 5x - 14 > 0$. **(4 marks)**

(E) **7** Find the set of values of $x$ for which

**a** $2(3x - 1) < 4 - 3x$ **(2 marks)**

**b** $2x^2 - 5x - 3 < 0$ **(4 marks)**

**c** **both** $2(3x - 1) < 4 - 3x$ **and** $2x^2 - 5x - 3 < 0$. **(2 marks)**

(E/P) **8** Given that $x \neq 3$, find the set of values for which $\frac{5}{x - 3} < 2$. **(6 marks)**

**Problem-solving**

Multiply both sides of the inequality by $(x - 3)^2$.

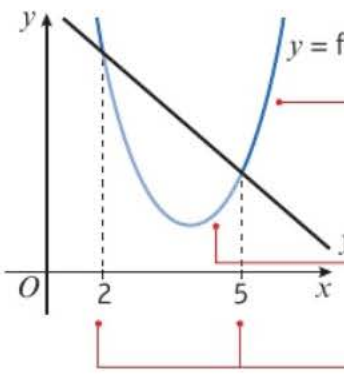
(E/P) **9** The equation $kx^2 - 2kx + 3 = 0$, where $k$ is a constant, has no real roots.

Prove that $k$ satisfies the inequality $0 \leqslant k < 3$. **(4 marks)**

## 3.6 Inequalities on graphs

You may be asked to interpret graphically the solutions to inequalities by considering the graphs of functions that are related to them.

- **The values of $x$ for which the curve $y = f(x)$ is below the curve $y = g(x)$ satisfy the inequality $f(x) < g(x)$.**
- **The values of $x$ for which the curve $y = f(x)$ is above the curve $y = g(x)$ satisfy the inequality $f(x) > g(x)$.**




f($x$) is above g($x$) when $x < 2$ and when $x > 5$.
These values of $x$ satisfy f($x$) > g($x$).

f($x$) is below g($x$) when $2 < x < 5$. These values of $x$ satisfy f($x$) < g($x$).

The solutions to f($x$) = g($x$) are $x = 2$ and $x = 5$.

## Example 12

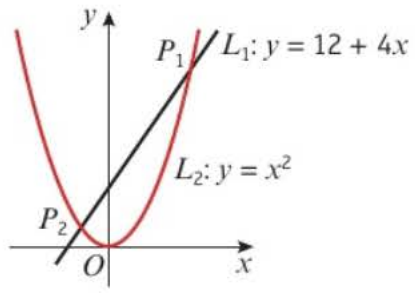
$L_1$ has equation $y = 12 + 4x$.

$L_2$ has equation $y = x^2$.

The diagram shows a sketch of $L_1$ and $L_2$ on the same axes.

**a** Find the coordinates of $P_1$ and $P_2$, the points of intersection.

**b** Hence write down the solution to the inequality $12 + 4x > x^2$.




**a** $x^2 = 12 + 4x$

$x^2 - 4x - 12 = 0$

$(x - 6)(x + 2) = 0$

$x = 6$ and $x = -2$

substitute into $y = x^2$

when $x = 6$, $y = 36$ $P_1$ (6, 36)

when $x = -2$, $y = 4$ $P_2$ (−2, 4)

**b** $12 + 4x > x^2$ when the graph of $L_1$ is above the graph of $L_2$

$-2 < x < 6$

Equate to find the points of intersection, then rearrange to solve the quadratic equation.

Factorise to find the $x$-coordinates at the points of intersection.

This is the range of values of $x$ for which the graph of $y = 12 + 4x$ is above the graph of $y = x^2$ i.e. between the two points of intersection.

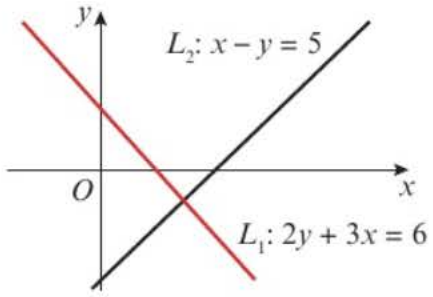
In set notation this is $\{x : -2 < x < 6\}$.

## Exercise 3F

**1** $L_1$ has equation $2y + 3x = 6$.
$L_2$ has the equation $x - y = 5$.
The diagram shows a sketch of $L_1$ and $L_2$.

**a** Find the coordinates of $P$, the point of intersection.

**b** Hence write down the solution to the inequality $2y + 3x > x - y$.



**2** For each pair of functions:

**i** Sketch the graphs of $y = f(x)$ and $y = g(x)$ on the same axes.

**ii** Find the coordinates of any points of intersection.

**iii** Write down the solutions to the inequality $f(x) \leqslant g(x)$.

**a** $f(x) = 3x - 7$
$g(x) = 13 - 2x$

**b** $f(x) = 8 - 5x$
$g(x) = 14 - 3x$

**c** $f(x) = x^2 + 5$
$g(x) = 5 - 2x$

**d** $f(x) = 3 - x^2$
$g(x) = 2x - 12$

**e** $f(x) = x^2 - 5$
$g(x) = 7x + 13$

**f** $f(x) = 7 - x^2$
$g(x) = 2x - 8$

(P) **3** Find the set of values of $x$ for which the curve with equation $y = f(x)$ is below the line with equation $y = g(x)$.

**a** $f(x) = 3x^2 - 2x - 1$
$g(x) = x + 5$

**b** $f(x) = 2x^2 - 4x + 1$
$g(x) = 3x - 2$

**c** $f(x) = 5x - 2x^2 - 4$
$g(x) = -2x - 1$

**d** $f(x) = \frac{2}{x}, x \neq 0$
$g(x) = 1$

**e** $f(x) = \frac{3}{x^2} - \frac{4}{x}, x \neq 0$
$g(x) = -1$

**f** $f(x) = \frac{2}{x+1}, x \neq -1$
$g(x) = 8$

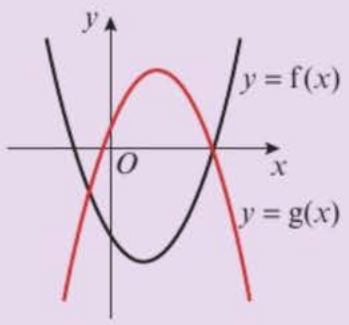
**Challenge**

The sketch shows the graphs of

$f(x) = x^2 - 4x - 12$
$g(x) = 6 + 5x - x^2$

**a** Find the coordinates of the points of intersection.

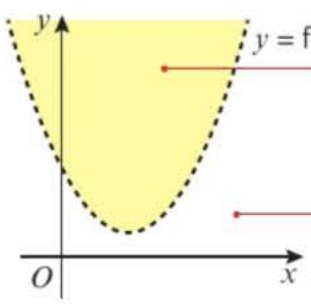
**b** Find the set of values of $x$ for which $f(x) < g(x)$.
Give your answer in set notation.




## 3.7 Regions

You can use shading on graphs to identify regions that satisfy linear and quadratic inequalities.

- **$y < f(x)$ represents the points on the coordinate grid below the curve $y = f(x)$.**
- **$y > f(x)$ represents the points on the coordinate grid above the curve $y = f(x)$.**




All the shaded points in this region satisfy the inequality $y > f(x)$.

All the unshaded points in this region satisfy the inequality $y < f(x)$.